

```
from google.colab import files
uploaded = files.upload()

import cv2
import numpy as np
import matplotlib.pyplot as plt

img = cv2.imread(list(uploaded.keys())[0])

img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)

sobel_x_abs = cv2.convertScaleAbs(sobel_x)
sobel_y_abs = cv2.convertScaleAbs(sobel_y)

sobel_xy = cv2.addWeighted(sobel_x_abs, 0.5, sobel_y_abs, 0.5, 0)

plt.figure(figsize=(14,5))

plt.subplot(1,4,1)
plt.title("Original Image")
plt.imshow(img_rgb)
plt.axis('off')

plt.subplot(1,4,2)
plt.title("Sobel X")
plt.imshow(sobel_x_abs, cmap='gray')
plt.axis('off')

plt.subplot(1,4,3)
plt.title("Sobel Y")
plt.imshow(sobel_y_abs, cmap='gray')
plt.axis('off')

plt.subplot(1,4,4)
plt.title("Sobel XY (Combined)")
plt.imshow(sobel_xy, cmap='gray')
plt.axis('off')

plt.show()
```

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Original Image



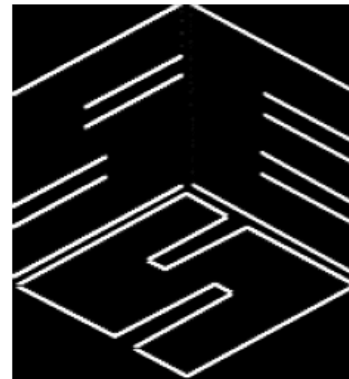
ENGINEER TO EXCEL

Sobel X



ENGINEER TO EXCEL

Sobel Y



ENGINEER TO EXCEL

Sobel XY (Combined)



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